Biometric Terminal — ODM RFQ / Sourcing Brief

Request for quotation · face + fingerprint access terminal · buyer runs its own firmware

> **Purpose.** We buy an integrated **face + fingerprint** access terminal and run **our own firmware** on it. We own the application and cloud layer; we need you to supply the biometric hardware + engine and a **Linux base we can root, reflash, and run fully offline**. Please answer every item in Sections 2 and 3 — those decide whether we can qualify your unit, not just its price.

1. Required SKU

MUST-HAVE

- **Face recognition**, on-device, **with liveness / anti-spoofing** (dual visible + IR camera).
- **Fingerprint** (capacitive or optical) with local matching.
- **Wi-Fi** (2.4 GHz minimum, dual-band preferred) **and Ethernet** (state if PoE).
- Baseline: display, speaker/mic, **door relay (dry contact)**, 12 V or PoE power, indoor mount.

QUOTE AS OPTIONS QR scanner, RFID/NFC card reader, Wiegand/RS485, outdoor/IP-rated enclosure, touch screen, battery/UPS.

NOT NEEDED palm-vein, vendor cloud/app integration, your management platform.

2. Firmware / OS requirements — the dealbreakers

We flash and maintain our own rootfs. A unit that fails any of these we cannot use at any price. Please answer each explicitly:

G1 **OS + kernel version** (e.g. Linux 4.x/5.x), libc (glibc/musl), and **CPU arch/ABI** (e.g. ARMv7 `arm`, ARM64 `aarch64`) so we can cross-compile.

G2 **Root access** on shipped units — serial-console pinout + how to obtain a root shell (documented). Required for evaluation.

G3 **Rootfs reflash path** — at least one of: a factory **USB flasher** (tool + **documented package format**) and/or **in-device partition reflash** (MTD/eMMC/UBI). Describe the **partition layout** (bootloader / kernel / dtb / rootfs / recovery / userdata).

G4 **Secure boot / signature verification** — is a **modified rootfs** allowed to boot? If signing is enforced, is there a documented way to enroll our own key?

G5 **Recovery / brick recovery** — is there a recovery image and a mask-ROM/USB recovery so a bad flash is recoverable?

G6 **Writable persistent partition** (`/data`-style) that **survives a rootfs reflash** (holds our config + enrolled templates).

G7 **Offline operation** — face match, fingerprint match, and door-open must work with **no internet and no vendor cloud**. Please confirm.

G8 **Golden firmware** — will you provide the **stock firmware package** (full image) as our restore baseline?

3. Biometric engine & data — cost, licensing, integration

1. **Face SDK** — which engine, and **per-device license cost**: is it **included** in the unit price or billed separately/recurring?
2. **Fingerprint** — sensor model + matching SDK; same licensing question.
3. **Local API** — how does an on-device app call enroll / match / liveness **without the vendor cloud** (IPC / C SDK / local HTTP)? We need this documented to integrate our firmware.
4. **Liveness method** — 2D-IR vs structured-light vs ToF, and the anti-spoof claim (printed photo / screen / mask). Anti-spoof strength matters for our use case.
5. **Template storage** — where templates live on-device, their **format**, and whether they can be **exported/imported** (we manage the roster server-side and sync enrollments across units).
6. **Camera spec** — visible + IR module, resolution, FOV, minimum illumination.

4. Hardware baseline (please fill in)

Item	Your spec
SoC + NPU (TOPS)	
RAM / flash (type, size)	
Display (size, resolution, touch?)	
Camera (visible + IR, resolution, liveness)	
Ethernet (speed, PoE?) / Wi-Fi (bands) / BLE?	
I/O (relay, Wiegand, RS485, GPIO)	
Power input	
Enclosure (material, IP rating, mount)	
Certifications (CE / FCC / RoHS; other)	

5. Commercial

- **Unit price at MOQ tiers:** 100 / 500 / 1,000 / 5,000.
- **NRE**, if any (we do our own UI/branding; mainly SKU/feature variants).
- **Lead time, warranty,** and **SoC/SKU longevity** (how long will you supply this exact unit).
- **Eval samples:** price + lead time for **2–3 units delivered with root access, the flashing tool, the partition-layout doc, the biometric local-API doc, and the golden firmware** (Sections 2–3).

6. What we do once we have a qualifying unit

Our firmware is a thin, portable layer: a few static ARM binaries + a small web UI + our backend integration. Per unit we only cross-compile for your ABI, adapt the boot/init hooks, reflash the rootfs, and call your biometric local API. We do **not** need you to build software — we need **root, a documented reflash path, no blocking secure boot, and a documented offline biometric API.** Those four are what we compare across quotes.

Please return answers on the accompanying response form so we can compare suppliers side by side. Thank you.